APPLIED ENGINEERING SCIENCES · MSU

# Jacob Johnson

Product design and mechanical engineering — working across CAD, CNC manufacturing, and code.




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CAD design in Siemens NX — not manufactured

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CAD, manufacturing, programming · MSU Infrastructure Planning and Facilities



WEBSITE
engineering-portfolio-sable.vercel.app

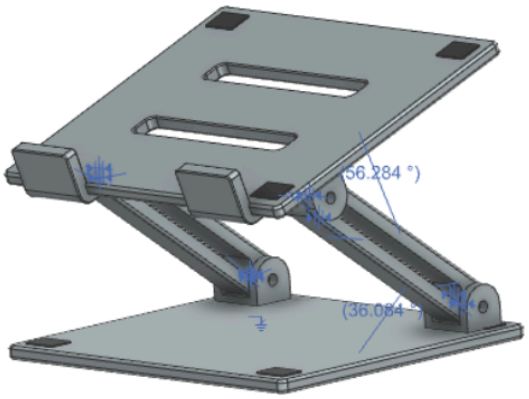
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CAD DESIGN ONLY · NOT MANUFACTURED

# Adjustable Laptop Stand

An ergonomic, height- and angle-adjustable laptop stand designed in Siemens NX, built around adjustable support arms, pivot joints, and a fully constrained assembly.




Full assembly in a mid-height working position (Siemens NX).

| TYPE | STATUS | TOOLS | SKILLS |
|---|---|---|---|
| Product Design · CAD | CAD design — not a manufactured prototype | Siemens NX | Parametric modeling, assembly constraints, mechanism design, ergonomics |

## OVERVIEW

Long hours on a laptop push the screen too low and the wrists into awkward positions. This project explores a stand that lets the user tune both the height and the viewing angle of the laptop to fit their own posture and workspace.

The design was developed entirely in Siemens NX as a digital product model: a base plate, a laptop platform, and a pair of support arms joined by pivots, brought together in a constrained assembly that can be moved through its range of positions.

## THE CHALLENGE

**Design a stand that is genuinely adjustable while still feeling stable and simple to use — the mechanism should add flexibility without adding complexity for the user.**

## DESIGN & PROCESS

01

### Adjustable support arms

Two support arms carry the laptop platform. Changing their angle raises, lowers, and tilts the platform, so one product can serve different users, desks, and tasks.

02

### Pivot joints

Clevis-style brackets on the base plate and the underside of the platform form pivot joints at both ends of each arm, giving the stand its range of motion.

03

### Assembly constraints

Mating and angle constraints make the parts behave like a real mechanism — the arms rotate about their pivots, and the platform follows.

04

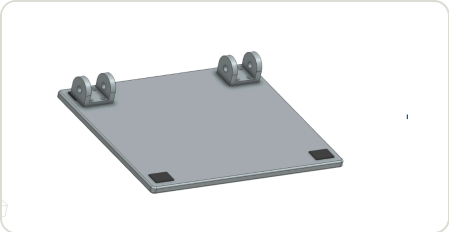
### Ergonomics & usability

Raising the screen toward eye level, angling the keyboard comfortably, and keeping the laptop secure with front lips and corner pads.

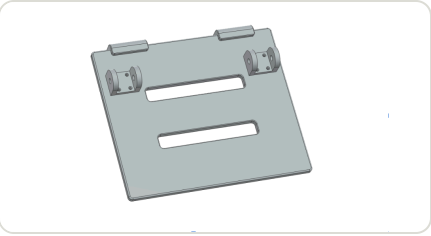
CAD DESIGN ONLY · NOT MANUFACTURED

# Components, range of motion, and joints

## COMPONENTS



Base plate with clevis brackets for the arm pivots

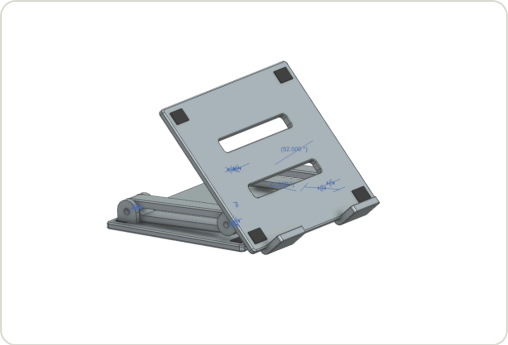


Platform underside with arm mounting brackets



Platform top with front lips and corner pads

## RANGE OF MOTION



Folded, low position

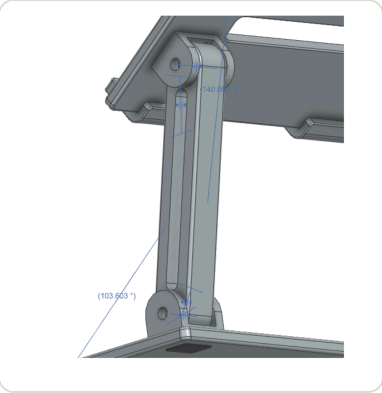


Raised position

## ARMS & PIVOT JOINTS



Rear view of the raised assembly



Support arm and pivot joint detail

## KEY FEATURES

- Two adjustable support arms with pivot joints at each end
- Base plate with clevis brackets and corner pads
- Platform with front retaining lips, corner pads, and cutouts
- Fully constrained multi-part assembly in Siemens NX

## RESULT

A complete Siemens NX part and assembly model. The constrained assembly can be posed from a folded position through mid-height and raised configurations, showing how the mechanism is intended to move.

**CAD design project.** It has not been manufactured or physically tested.

✓ PHYSICALLY MANUFACTURED · CNC-MACHINED & TESTED

# CNC-Machined Golf Club Face

A custom aluminum golf club face with pyramidal groove geometry, programmed in Mastercam, CNC-machined, and evaluated over 100 test shots.



The finished, machined club face

| | |
|---|---|
| TYPE | Manufacturing · CAM · Testing |
| STATUS | Machined prototype (non-USGA-conforming), range-tested |
| TIMELINE | April – May 2023 |
| TOOLS | Mastercam, G-code, CNC vertical mill |
| SKILLS | CAM programming, toolpath planning, dimensional verification, comparative testing |

## OVERVIEW

The grooves on a golf club face influence how much spin the ball carries and how it behaves after it lands. This project asked: what happens if the conventional groove pattern is replaced with a pyramidal geometry?

The face was designed to reduce ball spin and produce a flatter trajectory, programmed in Mastercam, converted to G-code, and machined from aluminum on a CNC vertical mill — then taken to the range and hit.

## THE CHALLENGE

**Produce an accurate, repeatable pyramidal surface pattern on a small aluminum part, then evaluate whether it changed the ball's behavior in a way that could be observed.**

## DESIGN & PROCESS

01

### Pyramidal geometry

Instead of straight grooves, the face is covered in a grid of small pyramids, intended to reduce ball spin and produce a flatter trajectory.

02

### Mastercam toolpaths

2.5D roughing and finishing toolpaths produce the face profile and the pyramid pattern from aluminum stock.

03

### CNC machining

Post-processed to G-code and run on a CNC vertical mill; the finished part was dimensionally verified against the CAD design.

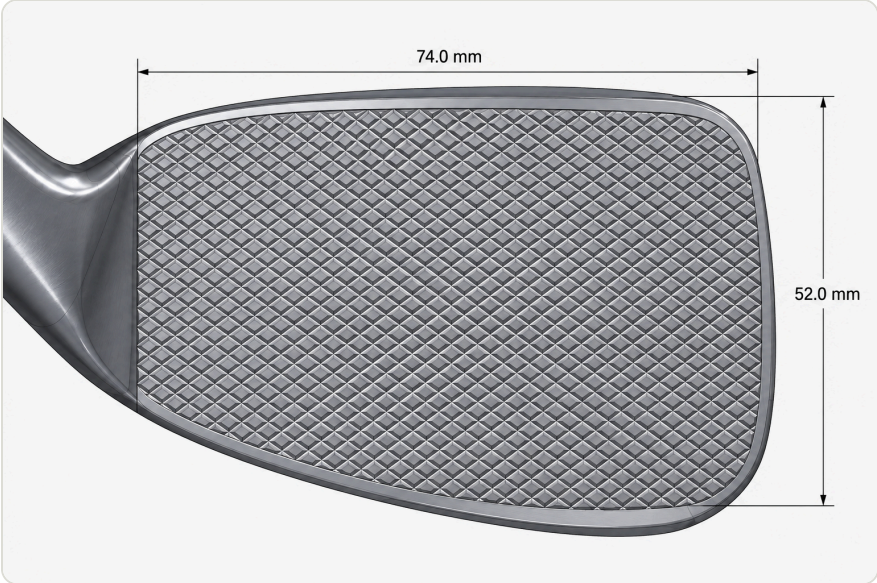
04

### Range testing

Evaluated over 100 test shots in comparative range testing against a standard club face.

PHYSICALLY MANUFACTURED · CNC-MACHINED & TESTED

# Design intent, realized in aluminum




Example illustration of the pyramidal face design (AI-generated)



Machined face, straight on

## KEY FEATURES

- Custom aluminum club face
- Pyramidal groove pattern across the full face
- 2.5D roughing and finishing toolpaths in Mastercam
- Machined on a CNC vertical mill and checked against the CAD model
- Evaluated over 100 test shots against a standard club face

## WHAT THE TESTING SHOWED

Across 100 test shots, I observed substantially reduced post-landing roll on most shots compared with a standard club face.

**Note:** Results are summarized qualitatively; no specific measured values are claimed. The face is a non-USGA-conforming prototype built for experimentation.

## WHAT I LEARNED

- Going from a design intent all the way to a physical part through CAM and G-code
- How toolpath choices shape what geometry is actually achievable
- Designing a simple comparative test and being honest about what it can and can't show

# Technical toolkit

| | |
|---|---|
| CAD | Siemens NX · Creo · SolidWorks · Fusion 360 · AutoCAD · Revit |
| MANUFACTURING | Mastercam · CNC toolpath planning · G-code |
| PROGRAMMING | Python · C · Java · AI-assisted development (Claude Code) |
| FIELD & DOCUMENTATION | Field measurements · AutoCAD drawing updates · Revit model updates |

EXPERIENCE

## MSU Infrastructure Planning and Facilities

DEC 2023 - JUN 2025

Technical Engineering Assistant

- Worked alongside engineers in MSU's facilities organization
- Created and revised AutoCAD drawings
- Updated Revit building models
- Conducted field measurements to verify and document existing conditions

WHAT I BRING

### Work ethic

I show up prepared and follow through. Working with professional engineers has taught me that accuracy and reliability matter as much as technical skill.

### Adaptability

My work spans parametric CAD, building information models, CNC programming, and software. I'm comfortable switching contexts and tools to fit the problem.

### Rapid learning

I learn new technical tools quickly — across six CAD and CAM platforms, three programming languages, and AI-assisted development with Claude Code.

CONTACT

# See the full interactive portfolio

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EDUCATION
Applied Engineering Sciences, MSU · May 2027